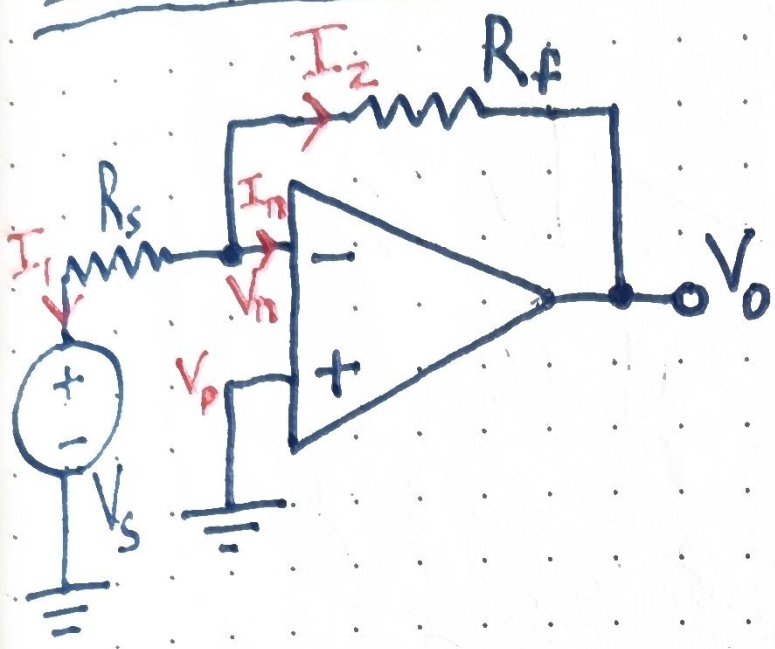
250929 Resistive OP-amps

Inverting Amp:



→ KCL @ V_n:

$$-I_1 - I_2 - \cancel{I_n}^{\,0} = 0$$

$$\Rightarrow I_1 = -I_2$$

→ From $V_n = V_p = 0$

$$\Rightarrow V_n = 0$$

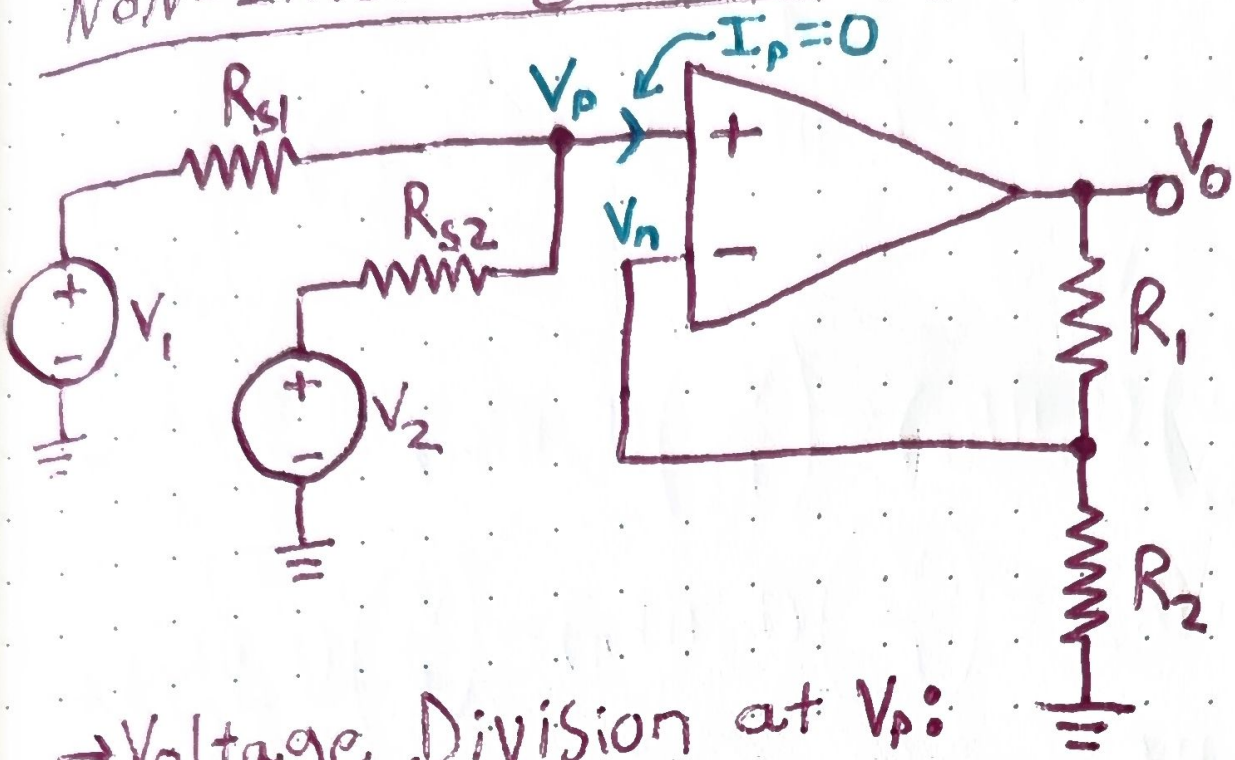
→ Ohm's Law:

$$I_1 = \frac{\cancel{V_n}^{\,0} - V_s}{R_s} \qquad I_2 = \frac{\cancel{V_n}^{\,0} - V_o}{R_f}$$

$$\Rightarrow I_1 = -I_2 \Rightarrow \frac{-V_s}{R_s} = \frac{V_o}{R_f}$$

$$\therefore \boxed{\frac{-R_f}{R_s} = \frac{V_o}{V_s} = G}$$

Non-Inverting Summing:



→ Voltage Division at V_p:

$$V_{RS2} = \frac{R_{S2}}{R_{S1} + R_{S2}} (V_1 - V_2) \quad \& \quad V_p = V_2 + V_{RS2}$$

$$\Rightarrow V_p = V_2 + \frac{R_{S2}}{R_{S1} + R_{S2}} (V_1 - V_2)$$

→ Voltage Division at V_n:

$$V_n = \frac{R_2}{R_1 + R_2} V_o$$

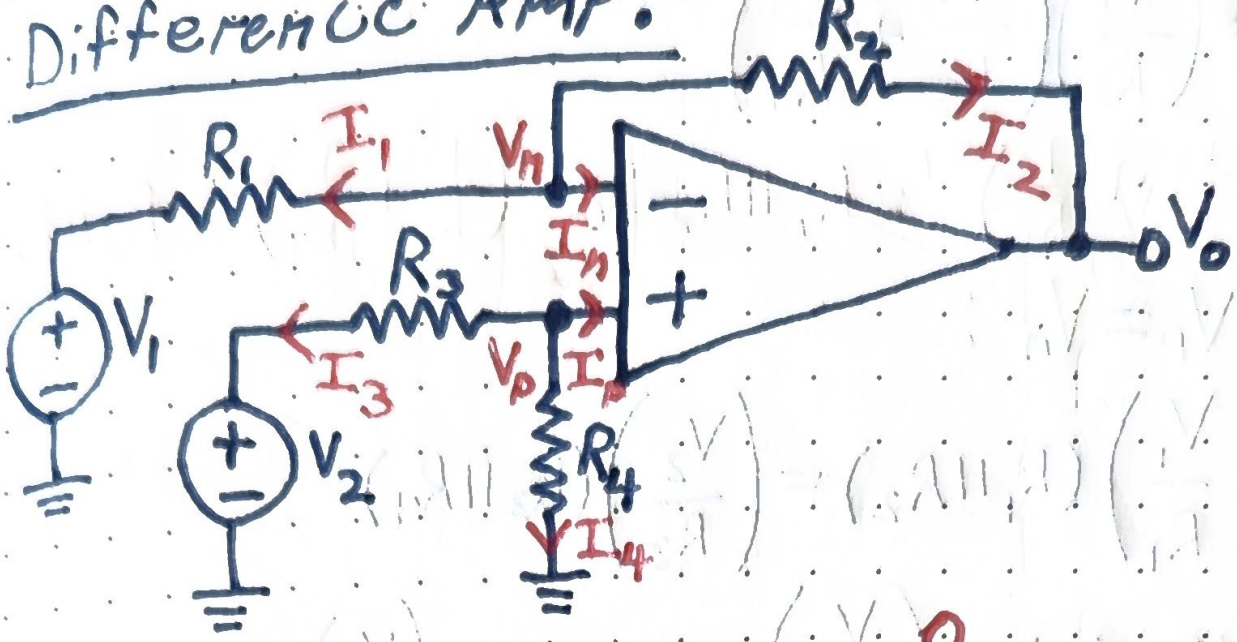
→ Use $V_p = V_n$:

$$\frac{R_2}{R_1 + R_2} V_o = V_2 + \frac{R_{S2}}{R_{S1} + R_{S2}} V_1 - \frac{R_{S2}}{R_{S1} + R_{S2}} V_2$$

Useing $1 - \frac{b}{a+b} = \frac{a}{a+b}$ If $a > 0$ and $b > 0$

$$\therefore \boxed{V_o = \left(\frac{R_1 + R_2}{R_2}\right)\left(\frac{R_{S2}}{R_{S1} + R_{S2}} V_1 + \frac{R_{S1}}{R_{S1} + R_{S2}} V_2\right)}$$

Difference Amp:



→ KCL @ V_n: $-I_1 - I_2 - \cancel{I_n}^{\,0} = 0 \Rightarrow I_1 = -I_2$

→ KCL @ V_p: $-I_3 - I_4 - \cancel{I_p}^{\,0} = 0 \Rightarrow I_3 = -I_4$

→ Use Ohm's Law:

$$I_1 = \frac{V_n - V_1}{R_1} \qquad I_2 = \frac{V_n - V_o}{R_2}$$

$$I_3 = \frac{V_p - V_2}{R_3} \qquad I_4 = \frac{V_p - 0}{R_4}$$

→ Use $I_1 = -I_2$ and solve for V_n:

$$\frac{V_n - V_1}{R_1} = \frac{V_o - V_n}{R_2} \Rightarrow \frac{V_n}{R_1} + \frac{V_n}{R_2} = \frac{V_1}{R_1} + \frac{V_o}{R_2}$$

$$\Rightarrow V_n = \left(\frac{V_1}{R_1} + \frac{V_o}{R_2}\right)\left(\frac{1}{R_1} + \frac{1}{R_2}\right)^{-1}$$

→ Use $I_3 = -I_4$ and solve for V_p:

$$\frac{V_p - V_2}{R_3} = \frac{-V_p}{R_4} \Rightarrow \frac{V_p}{R_3} + \frac{V_p}{R_4} = \frac{V_2}{R_3}$$

$$\Rightarrow V_p = \left(\frac{V_2}{R_3}\right)\left(\frac{1}{R_3} + \frac{1}{R_4}\right)^{-1}$$

let: $\left(\frac{1}{R_1} + \frac{1}{R_2}\right)^{-1} = R_1 \| R_2$ & $\left(\frac{1}{R_3} + \frac{1}{R_4}\right)^{-1} = R_3 \| R_4$

→ Use $V_p = V_n$!

$$\left(\frac{V_0}{R_2} + \frac{V_1}{R_1}\right)(R_1 \| R_2) = \left(\frac{V_2}{R_3}\right)(R_3 \| R_4)$$

$$\Rightarrow \left(\frac{V_0}{R_2}\right)(R_1 \| R_2) = \left(\frac{V_2}{R_3}\right)(R_3 \| R_4) - \left(\frac{V_1}{R_1}\right)(R_1 \| R_2)$$

Note: $R_1 \| R_2 = \frac{R_1 R_2}{R_1 + R_2}$ & $R_3 \| R_4 = \frac{R_3 R_4}{R_3 + R_4}$

$$\Rightarrow V_0\left(\frac{R_1}{R_1 + R_2}\right) = V_2\left(\frac{R_4}{R_3 + R_4}\right) - V_1\left(\frac{R_2}{R_1 + R_2}\right)$$

$$\therefore \boxed{V_0 = V_2\left(\frac{R_4}{R_3 + R_4}\right)\left(\frac{R_1 + R_2}{R_1}\right) - V_1\left(\frac{R_2}{R_1}\right)}$$